

Limits

Limit: If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a , (but not equal to a) then we write:

$$\lim_{x \rightarrow a} f(x) = L.$$

Example: $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

x	$f(x)$	x	$f(x)$
2.99	5.99	3.01	6.01
2.999	5.999	3.001	6.001
2.99999	5.99999	3.00001	6.00001

Here, we observe that the function $f(x)$ is undefined at $x=3$. But when we move closer to 3 the value of the function moves around 6. That means the limiting value of the function is 6 when the value of x moves toward 3.

Example-1: $\lim_{x \rightarrow 5} (x^2 - 4x + 3)$

$$= \lim_{x \rightarrow 5} (x^2) - \lim_{x \rightarrow 5} (4x) + \lim_{x \rightarrow 5} (3)$$

$$= 5^2 - (4 \times 5) + 3 = 8$$

∴ The value of the function approaches to 8 when x approaches to 5.

Problem-2: $\lim_{x \rightarrow 2} \frac{5x^3 + 4}{x - 3}$

$$= \lim_{x \rightarrow 2} \frac{5x^3 + 4}{x - 3}$$

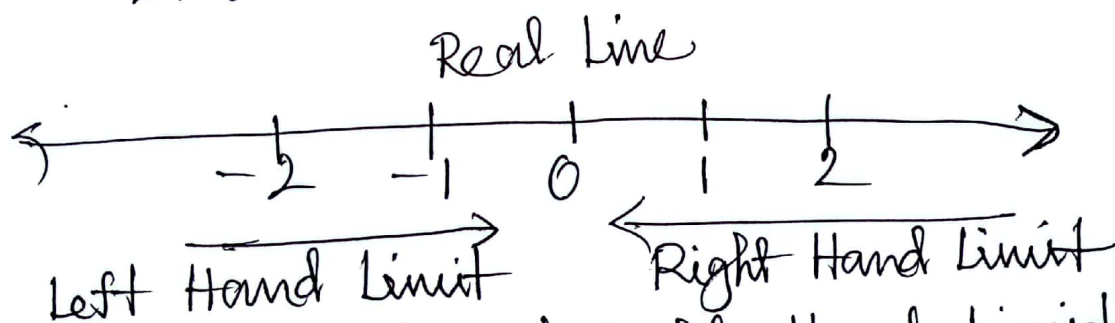
$$= \frac{5 \cdot 2^3 + 4}{2 - 3} = \frac{44}{-1} = -44.$$

Left Hand Limits: If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (less than a), then we write:

$$\lim_{x \rightarrow a^-} f(x) = L.$$

Right Hand Limits: If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (greater than a) then we write:

$$\lim_{x \rightarrow a^+} f(x) = L.$$



* These Left-Hand and Right-Hand Limits are often called as sided limits.

* Relationship Between One-sided and Two-sided Limits:

The two sided limit of function $f(x)$ exists at 'a' if and only if both of the one sided limit exists at 'a' and have the same value i.e.,

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

Example: Explain why $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

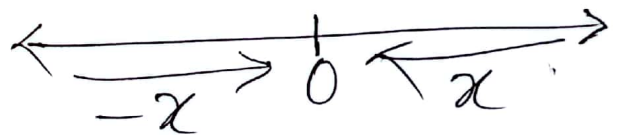
Solution: Let's assume that: $f(x) = \frac{|x|}{x}$.

Let's find out the left hand limits and right hand limit at $x=0$.

To find out the left hand limit we set:

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{|x|}{x} \\ &= \lim_{x \rightarrow 0^-} \frac{-x}{x} \\ &= \lim_{x \rightarrow 0^-} (-1) = -1 \end{aligned}$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



and right hand limit we set:

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} \\ &= \lim_{x \rightarrow 0^+} (1) = 1 \end{aligned}$$

Since Left Hand Limit $\neq$ Right Hand Limit (R.H.L.)

$\therefore \lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Example: A function $f(x)$ is defined as :

$$f(x) = \begin{cases} x-1 & \text{when } x \leq 3 \\ 3x-7 & \text{when } x > 3. \end{cases}$$

Find: a) $\lim_{x \rightarrow 3} f(x)$.

Solution: Given that:

$$f(x) = \begin{cases} x-1 & \text{when } x \leq 3 \\ 3x-7 & \text{when } x > 3. \end{cases}$$

As x approaches 3 from the right:

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (3x-7) = (3 \cdot 3 - 7) = 2$$

As x approaches 3 from the left:

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x-1) = 3-1 = 2.$$

Here, we observe that the left hand limit (L.H.L.) and the right hand limit (R.H.L.) exist and they are equal.

$$\therefore \lim_{x \rightarrow 3} f(x) = 2.$$

Example: A function $g(t)$ is defined as:

$$g(t) = \begin{cases} t-2 & \text{when } t < 0 \\ t^2 & \text{when } 0 \leq t \leq 2 \\ 2-t & \text{when } t > 2 \end{cases}$$

Find: (a) $\lim_{t \rightarrow 0} g(t)$; (b) $\lim_{t \rightarrow 1} g(t)$; (c) $\lim_{t \rightarrow 2} g(t)$.

Solution: Given that:

$$g(t) = \begin{cases} t-2 & \text{when } t < 0 \\ t^2 & \text{when } 0 \leq t \leq 2 \\ 2-t & \text{when } t > 2 \end{cases}$$

(a) $\lim_{t \rightarrow 0} g(t)$

As t approaches 0 from the left:

$$\lim_{t \rightarrow 0^-} g(t) = \lim_{t \rightarrow 0^-} (t-2) = -2.$$

As t approaches from the right,

$$\lim_{t \rightarrow 0^+} g(t) = \lim_{t \rightarrow 0^+} (t^2) = 0.$$

The left hand and right hand limit exist but they are not equal.

$\therefore \lim_{t \rightarrow 0} g(t)$ does not exist.

(b) $\lim_{t \rightarrow 1} g(t)$

When t approaches 1 from the left, the corresponding function is $g(t) = t^2$. Again when t approaches 1 from the right, the corresponding function is the same as before. So, the left hand limit (L.H.L.) and the right hand limit (R.H.L.) exist and they are equal. i.e.

$$\lim_{t \rightarrow 1} g(t) = \lim_{t \rightarrow 1} t^2 = 1.$$

(c) $\lim_{t \rightarrow 2} g(t)$.

As t approaches 2 from the left:

$$\lim_{t \rightarrow 2^-} g(t) = \lim_{t \rightarrow 2^-} (t^2) = 2^2 = 4$$

As t approaches 2 from the right:

$$\lim_{t \rightarrow 2^+} g(t) = \lim_{t \rightarrow 2^+} (2t) = 4$$

As, L.H.L. = R.H.L.

$$\therefore \lim_{t \rightarrow 2} g(t) = 4.$$

'H.W.' A function $f(x)$ is defined: $f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5, & -2 \leq x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$

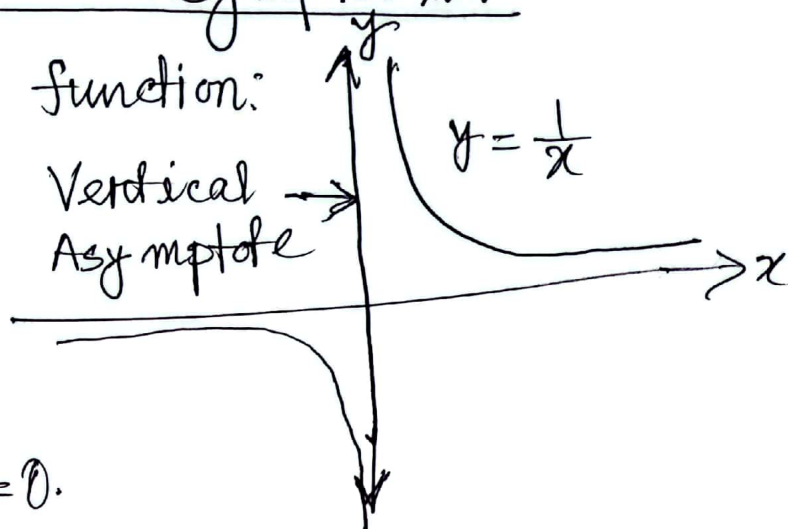
Find: (a) $\lim_{x \rightarrow 2} f(x)$ (b) $\lim_{x \rightarrow 0} f(x)$ (c) $\lim_{x \rightarrow 3} f(x)$.

*Horizontal and Vertical Asymptotes:

Let us consider the function:

$$y = f(x) = \frac{1}{x}$$

Now, we calculate the limit of the function $f(x)$ and $x=0$.



$$\text{Then } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

In each case, the function $f(x)$ either rises or falls without bound getting closer and closer to the vertical line $x=0$ as x approaches 0 from the sides indicated in the limit. The line $x=0$ is called vertical asymptote of the curve $y=f(x)$. Now, let's calculate the limit of the function $f(x)$ when x approaches to $+\infty$ and $-\infty$.

$$\text{Then, } \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \frac{1}{x} = 0.$$

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Here, $\lim_{x \rightarrow +\infty} f(x) = 0$ and $\lim_{x \rightarrow -\infty} f(x) = 0$.

$\therefore y=0$ is the horizontal asymptote of the curve of $f(x)$.

Example: Find the horizontal asymptote of $f(x) = \tan^{-1}x$.

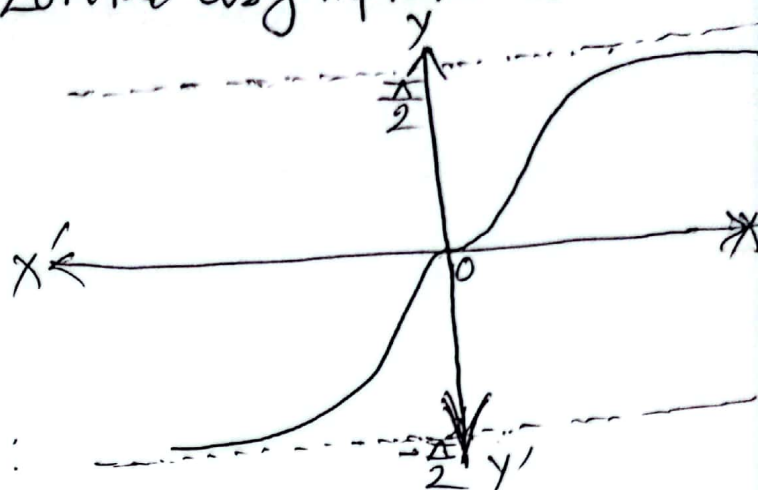
Solution: When x approaches $+\infty$:

$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \tan^{-1}(x) = \frac{\pi}{2}$$

When x approaches $-\infty$:

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \tan^{-1}(x) = -\frac{\pi}{2}$$

$\therefore y = \frac{\pi}{2}$ is a horizontal asymptote for f in the positive direction and the line $y = -\frac{\pi}{2}$ is a horizontal asymptote in the negative direction.



Limits of Polynomials:

$$\lim_{x \rightarrow 5} x^2 - 4x + 3 = 5^2 - 4 \times 5 + 3 = 8$$

The end behaviour of a polynomial matches the end behaviour of its highest degree term.

More precisely, if $c_n \neq 0$ then,

$$\lim_{x \rightarrow -\infty} (c_0 + c_1x + c_2x^2 + \dots + c_nx^n) = \lim_{x \rightarrow -\infty} c_nx^n$$

$$\lim_{x \rightarrow \infty} (c_0 + c_1x + c_2x^2 + \dots + c_nx^n) = \lim_{x \rightarrow \infty} c_nx^n$$

Ex: $\lim_{x \rightarrow -\infty} (7x^5 - 4x^3 + 2x - 9) = \lim_{x \rightarrow -\infty} (7x^5) = -\infty$

$$\lim_{x \rightarrow +\infty} (4x^8 + 17x^3 - 5x - 1) = \lim_{x \rightarrow \infty} 4x^8 = \infty$$

* Limits of Rational Functions:

Our technique for determining the end behaviour of a rational function is to divide each term in the numerator and denominator by the highest power of x that occurs in the denominator.

Example: Find $\lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8}$

We divide each term in the numerator and denominator by the highest power of x that occurs in the denominator namely $x^1 = x$. We obtain:

$$\lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8} = \lim_{x \rightarrow +\infty} \frac{3 + \frac{5}{x}}{6 - \frac{8}{x}}$$

$$= \frac{\lim_{x \rightarrow +\infty} (3 + \frac{5}{x})}{\lim_{x \rightarrow +\infty} (6 - \frac{8}{x})}$$

$$= \frac{\lim_{x \rightarrow +\infty} (3) + \lim_{x \rightarrow +\infty} (\frac{5}{x})}{\lim_{x \rightarrow +\infty} (6) - \lim_{x \rightarrow +\infty} (\frac{8}{x})}$$

$$= \frac{3 + 5 \lim_{x \rightarrow +\infty} \frac{1}{x}}{6 - 8 \lim_{x \rightarrow +\infty} \frac{1}{x}}$$

$$= \frac{3+0}{6-0} = \frac{3}{6} = \frac{1}{2} \quad \text{④}$$

H.W.: Find: (a) $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$ (b) $\lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$

Ans: (a) 0 . (b) $-\infty$

Example: Find: $\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+2}}{3x-6}$

It would be helpful to manipulate the function so that the powers of x are transformed to powers of $\frac{1}{x}$. This can be achieved by dividing the numerator and denominator by $|x|$ and using the fact that $\sqrt{x^2} = |x|$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+2}}{3x-6} &= \lim_{x \rightarrow +\infty} \frac{\frac{\sqrt{x^2+2}}{|x|}}{\frac{3x-6}{|x|}} = \lim_{x \rightarrow +\infty} \frac{\frac{\sqrt{x^2+2}}{\sqrt{x^2}}}{\frac{3x-6}{x}} \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{1+\frac{2}{x^2}}}{3-\frac{6}{x}} = \frac{\sqrt{\lim_{x \rightarrow +\infty} (1) + \lim_{x \rightarrow +\infty} (\frac{2}{x^2})}}{\lim_{x \rightarrow +\infty} (3) - \lim_{x \rightarrow +\infty} (\frac{6}{x})} \\ &= \frac{\sqrt{1+(2 \times 0)}}{3-(6 \times 0)} = \frac{1}{3} \end{aligned}$$

H.W.: Find $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{3x-6}$

L'Hospital Rule:

Some intermediate forms are of the following type:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, 0^0, \infty^0, 1^\infty$$

Applying L'Hospital Rule:

(1) Check that the limit of $\frac{f(x)}{g(x)}$ is an intermediate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

(2) Differentiate f and g separately.

(3) Find the limit of $\frac{f'(x)}{g'(x)}$. If this limit is

finite, $+\infty$ or $-\infty$, then it equals to limit of $\frac{f(x)}{g(x)}$

Example: Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Solution: $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ $\left[\frac{0}{0} \text{ form} \right]$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{1} \quad [L' \text{ Hospital Rule}]$$

$$= \frac{1}{1} = 1.$$

Example: Find $\lim_{x \rightarrow 0} \frac{\tan x}{x}$.

Solution: $\lim_{x \rightarrow 0} \frac{\tan x}{x}$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x}{1} \quad [L' \text{ Hospital Rule}]$$

$$= \frac{1}{1} = 1$$

Example: Find $\lim_{x \rightarrow +\infty} \frac{x}{e^x}$

Solution: $\lim_{x \rightarrow +\infty} \frac{x}{e^x}$ $\left[\frac{\infty}{\infty} \text{ form} \right]$

$$= \lim_{x \rightarrow +\infty} \frac{1}{e^x} = 0.$$

Example: Find $\lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8}$.

Solution: $\lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8}$ $\left[\frac{\infty}{\infty} \text{ form} \right]$

$$= \lim_{x \rightarrow +\infty} \frac{3x+5}{6x-8}$$

$$= \lim_{x \rightarrow +\infty} \frac{3}{6} = \frac{1}{2} \quad [L' \text{ Hospital Rule}]$$

Intermediate form of type $0 \cdot \infty$:

If we get the intermediate form of the type into $\frac{0}{0}$ or $\frac{\infty}{\infty}$ of type.

Example: Find $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \tan x) \cdot \sec 2x$.

Solution: $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \tan x) \cdot \sec 2x$ [$0 \cdot \infty$ form]

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(1 - \tan x)}{\frac{1}{\sec 2x}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{(1 - \tan x)}{\cos 2x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{-\sec^2 x}{-2 \sin 2x} \quad [L'Hospital Rule]$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sec^2 x}{2 \sin 2x} = \frac{2}{2} = 1$$

Intermediate Forms of Type 0^0 , ∞^0 , 1^∞ :

Intermediate forms of these type can sometimes be evaluated by first introducing a dependent variable.

$$y = [f(x)]^{g(x)}$$

$$\Rightarrow \ln(y) = g(x) \ln[f(x)].$$

Example: Find $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$.

Solution: Let, $y = (1 + \sin x)^{1/x}$.

$$\Rightarrow \ln(y) = \ln[(1 + \sin x)^{1/x}]$$

$$\Rightarrow \ln(y) = \frac{1}{x} \ln(1 + \sin x)$$

$$\Rightarrow \lim_{x \rightarrow 0} \ln(y) = \lim_{x \rightarrow 0} \frac{1}{x} \ln(1 + \sin x).$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+\sin x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{\cos x}{1+\sin x}}{1} = \frac{1}{1} = 1 \quad (\text{Ans})$$

$$\therefore \lim_{x \rightarrow 0} \ln y = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} e^{\ln y} = e^1$$

$$\Rightarrow \lim_{x \rightarrow 0} y e^{\ln} = e$$

$$\Rightarrow \lim_{x \rightarrow 0} y = e. \quad [e^{\ln} = 1]$$

$$\Rightarrow \lim_{x \rightarrow 0} (1+\sin x)^{1/x} = e.$$

H.W.: Find $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{7x}\right)^{3x+5}.$

Continuity:

A function f is said to be continuous at $x=c$ if the following conditions are satisfied: (1) $f(c)$ is defined

(2) $\lim_{x \rightarrow c} f(x)$ exists

(3) $\lim_{x \rightarrow c} f(x) = f(c)$

Example: Determine whether the following function is continuous at $x=3$.

$$f(x) = \begin{cases} x+2 & \text{when } -1 \leq x \leq 3 \\ 14-x^2 & \text{when } 3 \leq x \leq 5 \end{cases}$$

Definition: A function f is said to be continuous on a closed interval $[a, b]$ if the following conditions are satisfied:

1) f is continuous on the open interval (a, b) .

2) f is continuous from the right at a :

i.e.: $\lim_{x \rightarrow a^+} f(x) = f(a)$.

3) f is continuous from the left at b .

i.e.: $\lim_{x \rightarrow b^-} f(x) = f(b)$.

Example: A function is defined as $f(x) = \sqrt{x-2}$. Is the function continuous on the interval $[2, +\infty)$.

Solution: Given, $f(x) = \sqrt{x-2}$.

Let's consider the open interval $(2, +\infty)$.

Let, c be any point on $(2, +\infty)$.

The value of the function $f(x)$ at $x=c$.

$$f(c) = \sqrt{c-2}.$$

Here, $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sqrt{x-2} = \sqrt{c-2}$.

For any point $c \in (2, +\infty)$ the function is defined and is continuous. It is true for all points $c \in (2, +\infty)$. So, the function is continuous on the open interval $(2, +\infty)$.

As x approaches 2 from the right:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \sqrt{x-2} = 0.$$

$$\text{At } x=2, f(2) = 0.$$

$\therefore \lim_{x \rightarrow 2^+} f(x) = f(2)$. It means that the function

$f(x)$ is continuous from the right at $x=2$.

As x approaches $+\infty$ from the left, there will be no point for which the function is not continuous. The value of the function at the endpoint will approach $+\infty$.

So, we can say that the function $f(x)$ satisfies all the three conditions of continuity. Therefore, the function $f(x)$ is continuous on the interval $[2, \infty)$.

H.W.: What can you say about the continuity of the function $f(x) = \sqrt{9-x^2}$ on the closed interval $[-3, 3]$.

*Differentiability:

A function $f(x)$ is said to be differentiable at x_0 if: $\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$ exists.

The function f' is defined by the form:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

is called the derivative of f with respect to x . The domain f' consists of all x in the domain of f for which the limit exists.

Example: Use the limit definition to find the derivative of $f(x) = \sin x$ at $x = \frac{\pi}{2}$.

Solution: The derivative of the function using limit definition can be written as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin(x) (\cos h - 1) + \cos(x) \sin(h)}{h}$$

$$\begin{aligned}
 \therefore f'(x) &= \lim_{h \rightarrow 0} \sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \lim_{h \rightarrow 0} \left(\frac{\sin h}{h} \right) \\
 &= \sin(x) \times 0 + \cos(x) \times 1 \\
 &= \cos(x)
 \end{aligned}$$

At $x = \frac{\pi}{2}$; $f'\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$. (Ans)

H.W.: Find the derivative of $f(x) = \sqrt{x}$ at $x = 9$.

* A function $f(x)$ is not differentiable at:

(1) Corner Points: At corner point, the slope of the secant lines have different limits from the left and from the right and hence the two sided limits that define the derivative does not exist.

(2) Vertical tangency: At a point of vertical tangency the slope of the secant line approaches $+\infty$ or $-\infty$ from the left and from the right.

'Example': Prove that $f(x) = |x|$ is not differentiable at $x=0$.

Solution: The derivative of $f(x)$ is defined

by: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$\Rightarrow f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{|h| - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{|h|}{h}$$

Let's calculate the Left Hand Limit and Right Hand Limit.

The function $|h|$ is defined by:

$$|h| = \begin{cases} h & \text{when } h \geq 0 \\ -h & \text{when } h < 0 \end{cases}$$

$$\therefore L f'(0) = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} (-1) = -1$$

$$R f'(0) = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} (1) = 1$$

Since the Left Hand Limit and Right Hand Limit of $f'(0)$ are not equal, then

$\lim_{h \rightarrow 0} \frac{|h|}{h}$ does not exist.

$\therefore f(x) = |x|$ is not differentiable at $x=0$.

Problem: Show that: $f(x) = \begin{cases} x^2 + 2 & \text{when } x \leq 1 \\ x + 2 & \text{when } x > 1 \end{cases}$

is continuous but not differentiable at $x=1$.

Solution: Given: $f(x) = \begin{cases} x^2 + 2 & \text{when } x \leq 1 \\ x + 2 & \text{when } x > 1 \end{cases}$

At $x=1$, the value of the function is:

$$f(1) = 1^2 + 2 = 3.$$

Now let's see if $\lim_{x \rightarrow 1} f(x)$ exists.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 2) = 1 + 2 = 3.$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x + 2) = 1 + 2 = 3.$$

$$\therefore \lim_{x \rightarrow 1} f(x) \text{ exists and } \lim_{x \rightarrow 1} f(x) = 3.$$

We observe that the limiting value of the function as $x \rightarrow 1$ and the functional value at $x=1$ are equal.

$$\lim_{x \rightarrow 1} f(x) = 3 = f(1).$$

Since, the function $f(x)$ satisfies all the three conditions, the function $f(x)$ is continuous at $x=1$.

To show the differentiability of the function $f(x)$ at $x=1$, we set:

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

Since the behaviour of the function changes at $x=1$, we have to calculate the Left Hand limit (L.H.L.) and Right Hand Limit (R.H.L.) at $x=1$.

$$Lf'(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{(1+h)^2 + 2 - (1^2 + 2)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{2h + h^2}{h} = \lim_{h \rightarrow 0^-} (2 + h) = 2.$$

$$Rf'(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1 + h + 2 - 3}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

Since, $Lf'(1) \neq Rf'(1)$.

So, the function is not differentiable at $x=1$. Proved

* Techniques of Differentiation:

* Theorem: If the function f is differentiable at x_0 , then f is continuous at x_0 . If f is continuous at x_0 , then it can not be said that, the function is differentiable at x_0 .

* Rules of differentiation:

Sum Rule:

$$\frac{d}{dx} (A f(x) + B g(x)) = A \frac{df}{dx} + B \frac{dg}{dx}$$

Product Rule:

$$\frac{d}{dx} (f(x) g(x)) = f(x) \frac{dg(x)}{dx} + g(x) \frac{df(x)}{dx}$$

Quotient Rule:

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}{[g(x)]^2} \cdot [g(x) \neq 0]$$

* Example: Evaluate $\frac{dy}{dx}$ of $y = (4x^2 - 1)(7x^3 + x)$ by product rule of differentiation.

Solution: Given, $y = \frac{(4x^2 - 1)}{u} \cdot \frac{(7x^3 + x)}{v}$

$$\frac{dy}{dx} = (4x^2 - 1) \frac{d}{dx} (7x^3 + x) + (7x^3 + x) \frac{d}{dx} (4x^2 - 1)$$

$$= (4x^2 - 1) (21x^2 + 1) + (7x^3 + x) \cdot 8x$$

$$= 84x^4 - 21x^2 + 4x^2 - 1 + 56x^4 + 8x^2$$

$$= 140x^4 - 9x^2 - 1 \quad \underline{\text{(Ans)}}$$

Example: Evaluate $\frac{dy}{dx}$ of $y = \frac{x^3 + 2x^2 - 1}{x + 5}$ by quotient rule of differentiation.

Solution: Given, $y = \frac{x^3 + 2x^2 - 1}{x + 5}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x+5) \frac{d}{dx}(x^3 + 2x^2 - 1) - (x^3 + 2x^2 - 1) \frac{d}{dx}(x+5)}{(x+5)^2} \\ &= \frac{(x+5)(3x^2 + 4x) - (x^3 + 2x^2 - 1)}{(x+5)^2} \\ &= \frac{3x^3 + 15x^2 + 4x^2 + 20x - x^3 - 2x^2 - 1}{(x+5)^2} \\ &= \frac{2x^3 + 17x^2 + 20x + 1}{(x+5)^2}\end{aligned}$$

Practice Sheet:

Chapter 1.2 $\rightarrow$ 31, 37, 38, 39, 40.

Chapter ^{1.5}~~2.2~~ $\rightarrow$ 11-22, 29, 30

Chapter 2.2 $\rightarrow$ 9, 14, 46, 47-50.

Chapter 2.3 $\rightarrow$ 1-24.

Solution: Given that:

$$f(x) = \begin{cases} x+2 & \text{when } -1 \leq x < 3 \\ 14-x^2 & \text{when } 3 \leq x \leq 5. \end{cases}$$

The value of the function at $x=3$ is:

$$f(3) = 14 - 3^2 = 14 - 9 = 5.$$

Now see if $\lim_{x \rightarrow 3} f(x)$ exists.

As x approaches 3 from the left,

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x+2) = 3+2 = 5.$$

As x approaches 3 from the right,

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (14-x^2) = 14-3^2 = 5.$$

The Left Hand Limit (L.H.L.) and Right Hand Limit (R.H.L.) exist and they are equal.

$$\therefore \lim_{x \rightarrow 3} f(x) \text{ exists and } \lim_{x \rightarrow 3} f(x) = 5.$$

Hence we observe that: $\lim_{x \rightarrow 3} f(x) = f(3) = 5$.

Since, the function $f(x)$ satisfies all the three conditions of continuity, therefore, the function $f(x)$ is continuous at $x=3$.

H.W.: A function $g(x)$ is defined as:

$$g(x) = \begin{cases} 5-x & \text{when } -1 \leq x \leq 2 \\ x^2-1 & \text{when } 2 < x < 3 \end{cases}$$

Determine whether $g(x)$ is continuous at $x=2$.